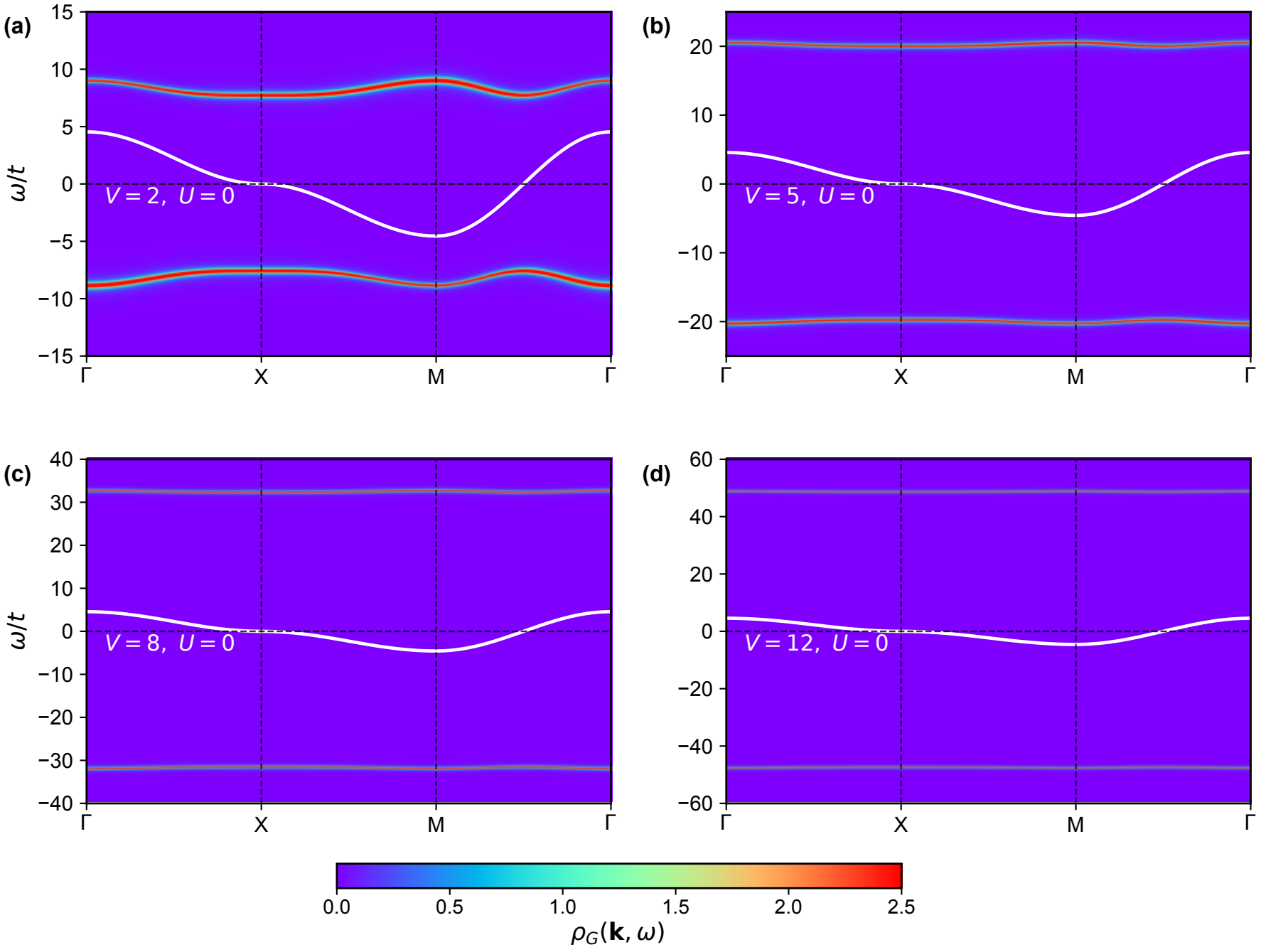

(a)
(b)
(c)
(d)
ω/t
V = 2, U = 0
V = 5, U = 0
V = 8, U = 0
V = 12, U = 0
Γ
X
M
Γ
15
10
5
0
−5
−10
−15
20
−20
40
30
−30
60
−40
−60
0.0
0.5
1.0
1.5
2.0
2.5
$\rho_G(\mathbf{k}, \omega)$